

MOHAMED MUSTAFA

Civil / Steel Structures Design Engineer

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PROFESSIONAL SUMMARY

Civil engineer specializing in steel structures design, with hands-on experience in API 650 above-ground storage tanks and 150+ delivered projects across the oil & gas, industrial, commercial and recreational sectors in Egypt and Saudi Arabia. Autodesk-certified in Revit, with strong hands-on command of SAP2000, ETABS, SAFE, STAAD.Pro, IDEA StatiCa, AutoCAD, Hilti and SPiT, plus advanced Excel-based design automation and a self-built LaTeX engine for calculation reports. Designs and checks to AISC 360, ASCE 7, API 650, SBC and ECP — from concept and analysis through connection design, shop-drawing review and on-site fabrication support.

CODES & STANDARDS

AISC 360 • AISC 341 • ASCE 7 • API 650 • SBC 301 / 306 • ECP 201 / 205

WORK EXPERIENCE

Design Engineer | *Soliman Company*

06/2024 – Present

Steel Structures

- Led the full steel structural design, from scratch, of heavy steel facilities for MBC Group (Saudi Arabia) and Organi Group (Egypt).
- Designed a heavy steel industrial factory for Al-Arjani Group, Egypt — 1,800 m² built-up area with a 30 m clear structural height, including full frame analysis, crane runway considerations and connection design.
- Delivered the steel design for petrol stations built from scratch: ExxonMobil (5+ stations), Total (3+ stations), Emirate E and AICCO, plus gas stations for Gas-Tech and Car-Gas.
- Designed structural steel caravans and portable units for Saudi Arabian projects, engineered for the governing wind and seismic loads per SBC 301 / ASCE 7 for the site zone and risk category.
- Developed advanced Excel design sheets fully compliant with Saudi (SBC), Egyptian (ECP) and American (AISC) codes, and programmed a LaTeX-based tool that auto-generates professional structural calculation notes.
- Coordinated technical work between multiple project sites and the head office, and reviewed shop drawings through fabrication and erection.

Storage Tanks — API 650

- ExxonMobil gasoline storage tank (Ø 9.0 m × 10.5 m height)** — full design and detailing to API 650: shell course thicknesses, bottom and annular plate layout, roof structure and plate layout, wind and seismic stability checks, anchorage and base design, with a complete calculation report and bill of materials.
- Total — Gasoline Storage Tank TK-08 (Ø 10.4 m × 13.57 m height), API 650 13th ed.** — design and detailing package covering the tank general arrangement, shell / bottom / roof plate layouts, shell and roof nozzles, shell and roof manholes, roof structure arrangement, foundation ring-beam interface, bill of materials and stamped design calculations.

Design Engineer (Part-Time) | *MAKLER Company*

10/2025 – Present

- Designed approximately 50 structural steel projects across the educational, industrial, commercial and recreational sectors.
- Key projects: the East Port Said National University Dome and the structural steel system of a water amusement park, alongside factories, warehouses and steel stairs.
- Prepared structural calculations, designed steel members and connections, and coordinated with the different engineering teams throughout each project.

Design Engineer (Part-Time) | *ORIENTALS Company*

06/2025 – 09/2025

- Supported the structural design and detailing and prepared the technical reports for the Al-Nahda Entertainment Complex (216,200 m²), Riyadh, Saudi Arabia.

EDUCATION

B.Sc. in Civil Engineering — Shoubra Faculty of Engineering, Benha University

- Graduation project: Steel Structures — Grade: Excellent. | Cumulative grade: Good (74.23%).

SELECTED ACHIEVEMENTS

- Delivered structural design for 150+ projects, including heavy steel for Organi Group, Carrefour, MBC Group and Porto Cruise, and light steel for Mobil, Total, Taqa, OLA Energy and Gas Master.
- Contributed to the structural steel design of Porto Cruise — New Alamein, the first ship-shaped (cruise-liner inspired) residential building in the Middle East (~57,000 m² landmark development), supporting the framing of its complex curved envelope.
- Extended the office's design capability into API 650 storage tanks, covering the shell, bottom, roof structure, internal floating roof, nozzles and manholes within a single in-house package.
- Designed bolted and welded connections, base plates and anchor bolts, and developed wind load calculations and load combinations for open and semi-open steel structures.
- Optimized steel tonnage while maintaining full structural safety and code compliance, and built the Excel-based design and automation tools now used across multiple projects.
- Designed steel canopies for petrol stations, including internal column drainage detailing, and provided ongoing technical support during fabrication and erection.

COURSES & CERTIFICATIONS

- Structural Revit Course — Autodesk Certified.
- General Safety and Risk Awareness — Total (French oil & gas corporation).
- CAD Master Diploma — Soliman Company. | English Language Course — Upper-Intermediate (B1).

TECHNICAL SKILLS

Analysis & Design	SAP2000, ETABS, SAFE, STAAD.Pro, IDEA StatiCa, Hilti PROFIS, SPiT
Drafting & BIM	AutoCAD, Revit (Autodesk Certified), Tekla Structures, Bluebeam Revu
Automation	Advanced Excel design sheets (VBA / formula-driven), LaTeX calculation-note generator
Office & Media	Word, Excel, PowerPoint — Excellent Photoshop, Premiere Pro — Very Good
Languages	Arabic — Native English — Very Good (B1)